I recommend building it as a **web-based Enrollment Analytics and Decision Support System** with four major layers:

1. **Data Collection & Management**
2. **Analytics & Visualization**
3. **AI/Prediction**
4. **Decision Support & Recommendations**

Below is a detailed roadmap you can use as your **master prompt for developing the system**, followed by the system architecture and diagrams.

1. Proposed System

System Title

Visualization of Enrollment Dynamics Utilizing Data Analytics with AI-Driven Decision Support

Main idea

The system will collect historical and current enrollment data and transform it into useful information such as:

- Enrollment trends by year/semester
- Enrollment by program
- Enrollment by year level
- Enrollment by gender, if appropriate and legally/ethically permitted
- Enrollment by department/college
- New vs. returning students
- Retention and dropout patterns
- Student population growth/decline
- Program demand
- Enrollment forecasting
- Identification of unusual enrollment changes
- AI-generated insights

- Recommendations for administrators

The important distinction is that the system should **not simply display graphs**. It should analyze the data and help administrators answer:

"What happened?" → "Why might it be happening?" → "What is likely to happen next?" → "What should we do?"

2. Master Prompt for Building the System

You can give the following prompt to an AI coding assistant and use it as your overall development specification.

MASTER DEVELOPMENT PROMPT

Project Title

Develop a complete web-based information system entitled:

"Visualization of Enrollment Dynamics Utilizing Data Analytics with AI-Driven Decision Support"

Project Objective

Build a secure, user-friendly, responsive, and data-driven enrollment analytics platform that allows educational administrators to monitor historical and current enrollment patterns, visualize enrollment dynamics, perform statistical and predictive analytics, and receive AI-assisted decision support.

The system must transform raw enrollment records into understandable dashboards, analytical reports, forecasts, alerts, and actionable recommendations.

The system should support the following decision-making cycle:

Data → Information → Analytics → Prediction → AI Insight → Recommendation → Decision

1. SYSTEM USERS

Implement role-based access control.

Administrator

The administrator can:

- Manage users
- Manage roles and permissions
- Import enrollment datasets
- Add/edit/delete enrollment records
- Manage academic years
- Manage semesters
- Manage colleges/departments
- Manage programs
- View all dashboards
- Generate reports
- Run forecasting
- View AI recommendations
- Configure system settings
- View system activity logs

Enrollment Officer / Registrar

The enrollment officer can:

- Upload enrollment datasets
- Validate records
- View enrollment dashboards
- Search/filter enrollment records
- Generate enrollment reports
- View trends

- View forecasts
- Export authorized reports

School Administrator / Decision Maker

The decision maker can:

- View dashboards
- Analyze enrollment trends
- Compare programs
- View forecasts
- Receive AI-generated insights
- View alerts
- Review recommended actions
- Generate management reports

Researcher / Analyst

The analyst can:

- Access approved datasets
- Perform statistical analysis
- Create visualizations
- Compare academic periods
- Generate analytical reports

---

2. CORE MODULES

Develop the system using the following modules.

Module A — Authentication and Authorization

Implement:

- Login
- Logout

- Password management
- Role-based access control
- Session management
- Account activation/deactivation
- Audit logging

Security requirements:

- Password hashing
- Input validation
- SQL injection protection
- CSRF protection where applicable
- Authentication middleware
- Authorization middleware
- Secure sessions
- Rate limiting for login attempts
- Audit logs

---

3. DATA MANAGEMENT MODULE

Create a centralized enrollment database.

The system should support:

Academic Information

- Academic year
- Semester
- Term
- College
- Department
- Program

- Major
- Year level

Enrollment Information

- Enrollment count
- New students
- Returning students
- Continuing students
- Transferees
- Graduating students
- Withdrawn students
- Dropped students
- Retained students

Optional demographic variables

Only include demographic fields that are institutionally necessary and permitted by applicable privacy policies.

Examples:

- Sex/gender
- Age group
- Residency category

Avoid collecting unnecessary personally identifiable information.

---

4. DATABASE DESIGN

Create normalized relational database tables.

Recommended tables:

users

- id

- name
- email
- password_hash
- role_id
- status
- created_at
- updated_at

roles

- id
- role_name
- description

academic_years

- id
- academic_year
- start_date
- end_date
- status

semesters

- id
- semester_name
- academic_year_id

colleges

- id
- college_name
- college_code

departments

- id
- department_name
- department_code
- college_id

programs

- id
- program_name
- program_code
- department_id
- degree_level
- status

enrollment_records

- id
- academic_year_id
- semester_id
- college_id
- department_id
- program_id
- year_level
- enrollment_type
- enrollment_count
- created_at
- updated_at

forecasts

- id
- program_id

- forecast_period
- predicted_enrollment
- lower_bound
- upper_bound
- model_used
- accuracy_metric
- created_at

ai_insights

- id
- insight_type
- title
- description
- severity
- supporting_metric
- recommendation
- created_at

audit_logs

- id
- user_id
- action
- module
- timestamp
- ip_address
- details

Use foreign keys and appropriate indexes.

---

5. DATA IMPORT MODULE

Allow authorized users to upload:

- CSV
- Excel
- JSON, if required

The import process must contain:

1. File upload
2. File validation
3. Column validation
4. Data-type validation
5. Missing-value detection
6. Duplicate detection
7. Invalid-value detection
8. Data preview
9. Validation report
10. User confirmation
11. Database insertion
12. Import log

Example validation:

If:

academic_year = 2025-2026
program = BSIT
semester = First Semester
year_level = 1
enrollment_count = 180

the system should verify that the record is valid before storing it.

Do not silently accept corrupted data.

---

6. DATA CLEANING

Implement automated data-quality checks.

Detect:

- Missing values
- Duplicate records
- Negative enrollment counts
- Invalid academic years
- Invalid programs
- Invalid year levels
- Inconsistent naming
- Impossible values
- Duplicate combinations of academic year + semester + program + year level

Display a data-quality score.

Example:

Data Quality Score: 96%

Issues:

- 12 missing values
- 4 duplicate records
- 2 invalid program codes

Allow authorized users to review and correct errors.

---

7. ANALYTICS ENGINE

Implement descriptive analytics.

Calculate:

Total Enrollment

Total number of enrolled students for the selected period.

Enrollment Growth

Use:

Growth Rate =

(Current Enrollment - Previous Enrollment)
/
Previous Enrollment × 100

Year-over-Year Growth

Compare enrollment between academic years.

Semester Growth

Compare semester-level enrollment.

Program Share

Calculate the percentage contribution of each program to total enrollment.

Retention Rate

Where appropriate:

Retention Rate =

Returning/Continuing Students
/
Eligible Students × 100

Attrition Rate

Where appropriate:

Attrition Rate =

Students Lost
/
Eligible Students × 100

Clearly document the institution's definitions for retention and attrition.

---

8. DASHBOARD

Create an interactive administrative dashboard.

The dashboard should contain:

KPI Cards

Display:

- Total Enrollment
- Enrollment Growth Rate
- Highest-Enrolled Program
- Lowest-Enrolled Program
- New Students
- Returning Students
- Retention Rate
- Forecasted Next-Period Enrollment

Filters

Provide:

- Academic year
- Semester
- College
- Department
- Program
- Year level
- Student type

Filters must update dashboard visualizations dynamically.

---

9. VISUALIZATION MODULE

Create professional interactive charts.

Chart 1 — Enrollment Trend

Line chart showing enrollment across academic years/semesters.

Chart 2 — Enrollment by Program

Bar chart ranking programs by enrollment.

Chart 3 — Enrollment by College

Bar chart comparing colleges.

Chart 4 — Program Distribution

Pie/donut chart showing enrollment share.

Chart 5 — Year-Level Distribution

Show enrollment from:

- First Year
- Second Year
- Third Year
- Fourth Year
- Other applicable levels

Chart 6 — New vs Returning Students

Stacked bar chart.

Chart 7 — Retention Trend

Line chart across academic periods.

Chart 8 — Enrollment Heatmap

Show:

Program × Academic Year

using enrollment intensity.

Chart 9 — Forecast

Display:

Historical Enrollment
+
Forecasted Enrollment

Include confidence/prediction intervals where supported.

---

10. INTERACTIVE VISUALIZATION REQUIREMENTS

Every visualization should support:

- Hover tooltips
- Filtering
- Drill-down
- Date/academic-period selection
- Program selection
- Export
- Responsive design

Clicking a college should allow the user to drill down into its departments.

Clicking a department should allow the user to drill down into its programs.

Example:

College
↓
Department
↓
Program
↓
Year Level
↓
Enrollment Trend

---

11. AI ANALYTICS MODULE

Implement an AI-assisted analytics layer.

The AI should NOT simply generate generic text.

It should receive verified analytical results from the analytics engine.

For example:

Input:

- Current enrollment = 4,500
- Previous enrollment = 4,850
- Growth = -7.2%
- BSIT growth = +14%
- BSBA growth = -18%

AI output:

"Overall enrollment declined by approximately 7.2%. However, BSIT increased while BSBA declined significantly. Administrators may investigate whether program-specific demand, capacity, scheduling, or recruitment factors contributed to this difference."

The AI should distinguish between:

FACT

Directly supported by data.

TREND

A statistically or visually observable pattern.

FORECAST

A model-generated estimate.

INTERPRETATION

Possible explanation.

RECOMMENDATION

Suggested action.

Never present speculation as fact.

---

12. AI INSIGHT GENERATION

Generate insights such as:

Trend Insight

"Enrollment has increased for three consecutive academic periods."

Decline Alert

"Enrollment in Program X decreased significantly compared with the previous period."

Growth Opportunity

"Program Y shows sustained enrollment growth and may warrant capacity assessment."

Capacity Alert

"Projected enrollment may exceed the current planning threshold."

Retention Alert

"Retention has declined across multiple periods."

Anomaly Alert

"Enrollment for this program is substantially different from its historical pattern."

---

13. PREDICTIVE ANALYTICS

Implement enrollment forecasting.

Possible models:

Baseline

- Moving average
- Naive forecast

Statistical models

- Linear regression
- Exponential smoothing
- ARIMA/SARIMA where appropriate

Machine-learning models

If sufficient historical data exists:

- Random Forest
- Gradient Boosting

- XGBoost
- Other appropriate regression models

Do not automatically use complex AI when simpler models perform better.

The system should compare models using appropriate validation metrics.

Possible metrics:

- MAE
- RMSE
- MAPE

Display model performance to administrators.

---

14. FORECASTING WORKFLOW

Implement:

Historical Data
↓
Data Cleaning
↓
Feature Engineering
↓
Train/Test Split
↓
Model Training
↓
Model Validation
↓
Model Comparison
↓
Best Model Selection
↓
Forecast Generation
↓
Confidence/Prediction Interval
↓

Visualization
↓
AI Interpretation

Forecast:

- Next semester
- Next academic year
- Program-level enrollment
- College-level enrollment

The forecast horizon should be configurable.

---

15. AI DECISION SUPPORT

Create a Decision Support Center.

For every major insight, display:

Situation

What is happening?

Evidence

What data supports it?

Possible Interpretation

What could explain the pattern?

Forecast

What may happen next?

Recommendation

What action could administrators consider?

Priority

- Low
- Medium
- High

- Critical

Confidence

Display an appropriate confidence indicator based on the underlying analysis/model.

Example:

Situation:
"Enrollment in BSCS has declined for two consecutive semesters."

Evidence:
"Enrollment decreased from 320 to 275 students."

Forecast:
"Model projects approximately 250–265 students next semester."

Recommendation:
"Review recruitment performance, student retention, scheduling constraints, and competitor/program demand indicators before allocating additional recruitment resources."

---

16. WHAT-IF ANALYSIS

Implement a scenario analysis feature.

Allow administrators to modify assumptions.

Example:

Current enrollment:
5,000

Scenario:

Recruitment improvement:
+10%

Retention improvement:
+5%

Capacity:
5,500

The system calculates an estimated scenario outcome.

Display:

Current Scenario
vs.
Improved Recruitment Scenario
vs.
Improved Retention Scenario

Clearly label scenario outputs as simulations rather than actual predictions.

---

17. ALERT SYSTEM

Create automated alerts.

Examples:

Enrollment decline

Trigger when enrollment falls below a configurable threshold.

Rapid growth

Trigger when enrollment exceeds a configurable growth threshold.

Retention decline

Trigger when retention falls below target.

Forecast capacity issue

Trigger when forecast exceeds capacity.

Data anomaly

Trigger when an observation is unusually different from historical patterns.

Alerts should contain:

- Alert title
- Date
- Affected program
- Metric
- Current value
- Expected/reference value

- Severity
- Explanation
- Recommended action

---

18. REPORT GENERATION

Allow administrators to generate reports.

Reports should include:

- Executive summary
- Enrollment overview
- Historical trends
- Program analysis
- College analysis
- Retention analysis
- Forecast
- AI-generated insights
- Recommendations
- Methodology
- Data quality information

Export formats:

- PDF
- Excel
- CSV

---

19. EXECUTIVE DASHBOARD

Create a simplified dashboard for decision makers.

Show only the most important information:

1. Current enrollment
2. Enrollment growth
3. Retention
4. Top programs
5. Declining programs
6. Forecast
7. Alerts
8. AI recommendations

Use visual indicators such as:

Green = positive
Yellow = needs attention
Red = critical

Do not use color as the only indicator; include text/icons for accessibility.

---

20. DATA PRIVACY

The system must protect student information.

Prefer aggregated data for analytics.

Do not expose personally identifiable student information in public dashboards.

Implement:

- Role-based access
- Authentication
- Authorization
- Encryption where appropriate
- Audit logging
- Secure file uploads
- Data minimization
- Access controls

- Backup procedures

AI should receive only the minimum information necessary for the requested analysis.

---

21. AI SAFETY AND GOVERNANCE

The AI must not make autonomous administrative decisions.

It should provide:

"Decision support"

rather than:

"Automatic decision making."

Every AI recommendation should be reviewable by an authorized administrator.

AI responses should include supporting metrics.

The system should avoid statements such as:

"Program X must be closed."

Instead use:

"Program X shows sustained enrollment decline. Administrators may consider reviewing program demand, retention, recruitment, and resource utilization before making any program-level decision."

---

22. SYSTEM ARCHITECTURE

Use a modular architecture.

Recommended structure:

Frontend
↓
REST API / Backend
↓
Authentication & Authorization
↓
Analytics Service
↓

AI Decision Support Service
↓
Forecasting Service
↓
Database

Use separate services/modules for:

- Authentication
- Data management
- Analytics
- Visualization
- Forecasting
- AI
- Reports
- Notifications
- Audit logging

---

23. API DESIGN

Create RESTful endpoints.

Example:

GET /api/enrollment

GET /api/enrollment/trends

GET /api/enrollment/programs

GET /api/enrollment/colleges

GET /api/enrollment/retention

GET /api/analytics/summary

POST /api/data/import

POST /api/forecast

GET /api/forecast

POST /api/ai/analyze

GET /api/ai/insights

GET /api/reports

GET /api/alerts

Use authentication and authorization for protected endpoints.

---

24. RECOMMENDED FRONTEND PAGES

Create:

1. Login

2. Dashboard

3. Enrollment Overview

4. Enrollment Trends

5. Program Analytics

6. College Analytics

7. Retention Analytics

8. Forecasting

9. AI Decision Support

10. Alerts

11. Data Import

12. Data Quality

13. Reports

14. User Management

15. System Settings

16. Audit Logs

---

25. UI/UX DESIGN

Use a clean academic/institutional dashboard design.

Recommended layout:

Sidebar:

- Dashboard
- Enrollment
- Analytics
- Programs
- Forecast
- AI Decision Support
- Alerts
- Reports
- Data Management
- Users
- Settings

Top bar:

- Search
- Academic year
- Notifications
- User profile

Main dashboard:

- KPI cards
- Trend charts
- Ranking charts
- Forecast
- AI insights

- Alerts

Design requirements:

- Responsive
- Desktop/tablet/mobile support
- Accessible
- Clear typography
- Consistent colors
- Minimal clutter
- Dark/light mode if appropriate

---

26. ANALYTICAL QUESTIONS THE SYSTEM MUST ANSWER

The system should allow administrators to answer:

1. How many students are enrolled?
2. Is enrollment increasing or decreasing?
3. Which programs are growing?
4. Which programs are declining?
5. Which college has the highest enrollment?
6. Which programs contribute the most students?
7. Which year level has the largest population?
8. What is the retention trend?
9. What programs show unusual changes?
10. What enrollment can we expect next semester?
11. What enrollment can we expect next academic year?
12. Which programs may require additional capacity?
13. Which programs may require recruitment attention?
14. What patterns are visible in historical data?

15. What possible actions should administrators consider?

---

27. DEVELOPMENT PHASES

Follow this order.

Phase 1 — Requirements

- Define users
- Define objectives
- Define KPIs
- Define data sources
- Define security requirements
- Define privacy requirements
- Define AI scope

Phase 2 — Database

- Create ERD
- Create tables
- Define relationships
- Create migrations
- Add indexes
- Add constraints

Phase 3 — Backend

- Authentication
- Authorization
- CRUD APIs
- Data import
- Validation
- Analytics APIs

Phase 4 — Frontend

- Login
- Navigation
- Dashboard
- Filters
- Tables
- Charts

Phase 5 — Analytics

- Descriptive statistics
- Growth calculations
- Retention
- Program ranking
- Trend analysis
- Anomaly detection

Phase 6 — Forecasting

- Prepare historical dataset
- Train baseline models
- Train candidate models
- Validate models
- Compare metrics
- Select model
- Generate forecast

Phase 7 — AI

- Build AI insight service
- Pass verified metrics to AI
- Generate explanations

- Generate recommendations
- Add confidence/context
- Prevent unsupported claims

Phase 8 — Decision Support

- Alerts
- Recommendations
- What-if analysis
- Executive dashboard

Phase 9 — Reports

- PDF
- Excel
- CSV
- Executive report

Phase 10 — Testing

Perform:

- Unit testing
- Integration testing
- API testing
- UI testing
- Security testing
- Data validation testing
- Forecast accuracy testing
- AI output evaluation
- User acceptance testing

Phase 11 — Deployment

Prepare:

- Production database
- Backend deployment
- Frontend deployment
- Environment variables
- HTTPS
- Database backups
- Logging
- Monitoring

---

28. TEST CASES

Create test cases for:

Authentication

- Valid login
- Invalid password
- Disabled account
- Unauthorized page access

Data Import

- Valid CSV
- Invalid CSV
- Missing columns
- Duplicate records
- Negative enrollment
- Empty dataset

Analytics

- Correct total enrollment
- Correct growth rate

- Correct ranking
- Correct retention calculation

Forecasting

- Historical dataset
- Insufficient data
- Missing periods
- Model failure
- Forecast generation

AI

- Correct interpretation
- Unsupported claim detection
- Recommendation generation
- Missing data handling

Security

- Unauthorized API access
- SQL injection
- Invalid file upload
- Session security

---

29. EXPECTED FINAL OUTPUT

The finished system should provide:

Input

Historical and current enrollment data.

Processing

Data cleaning
+
Statistical analytics

+
Visualization
+
Forecasting
+
AI analysis

Output

Interactive dashboards
+
Enrollment trends
+
Program analysis
+
Forecasts
+
Alerts
+
AI-generated insights
+
Decision recommendations
+
Reports

The final system should demonstrate that data analytics and AI can assist educational administrators in making evidence-based enrollment planning decisions.

---

30. IMPORTANT DEVELOPMENT RULE

Do not build the entire system at once.

Develop incrementally.

First build:

Database → Authentication → Data Import → Dashboard → Analytics → Forecasting → AI → Decision Support

After every phase:

1. Test it.

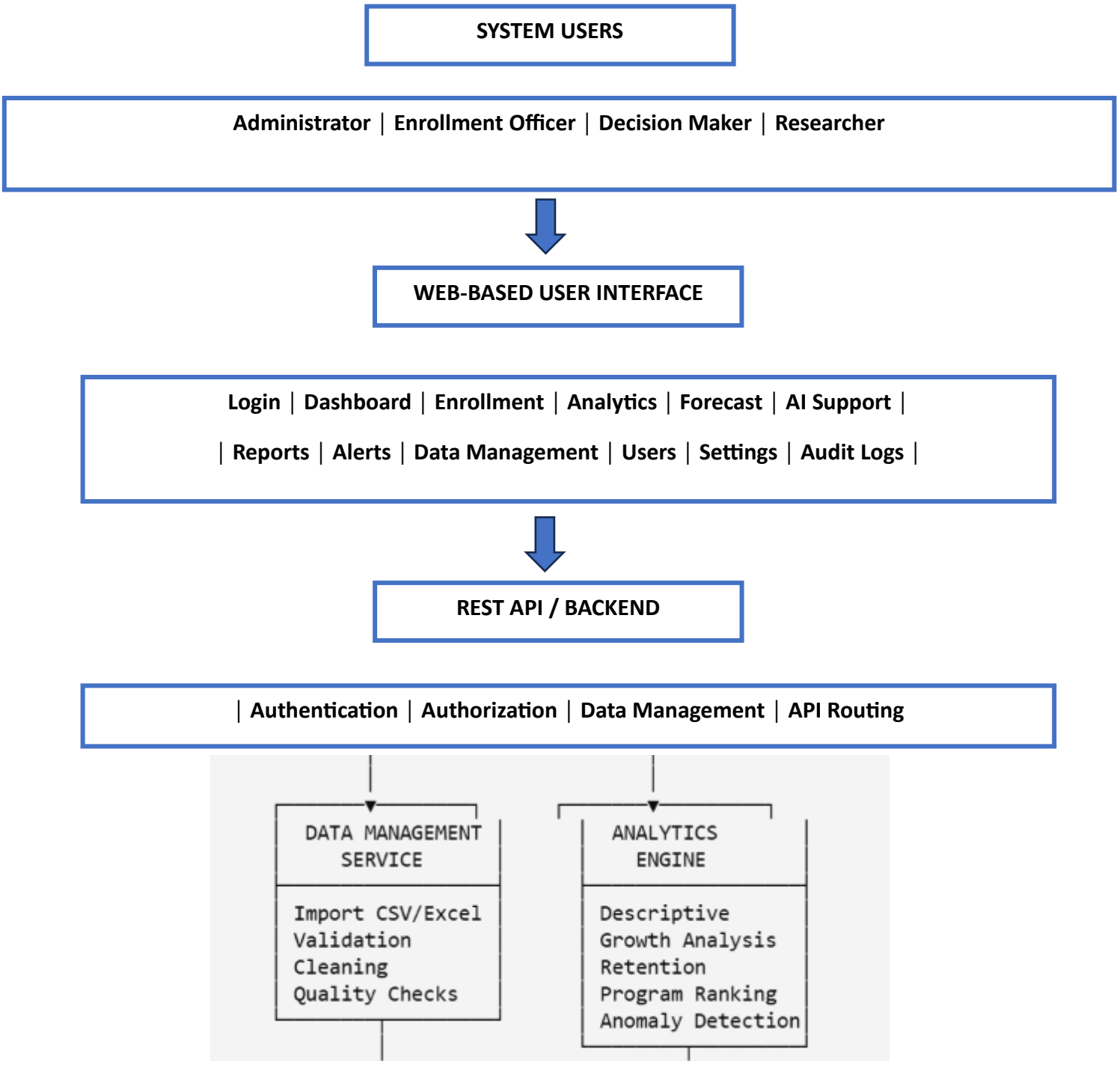
2. Fix errors.
3. Document it.
4. Commit the working version.
5. Continue to the next phase.

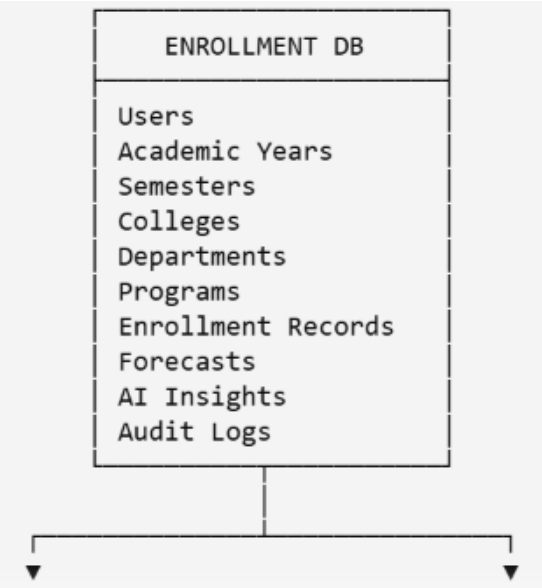
Do not implement AI before the underlying data and analytics are reliable.

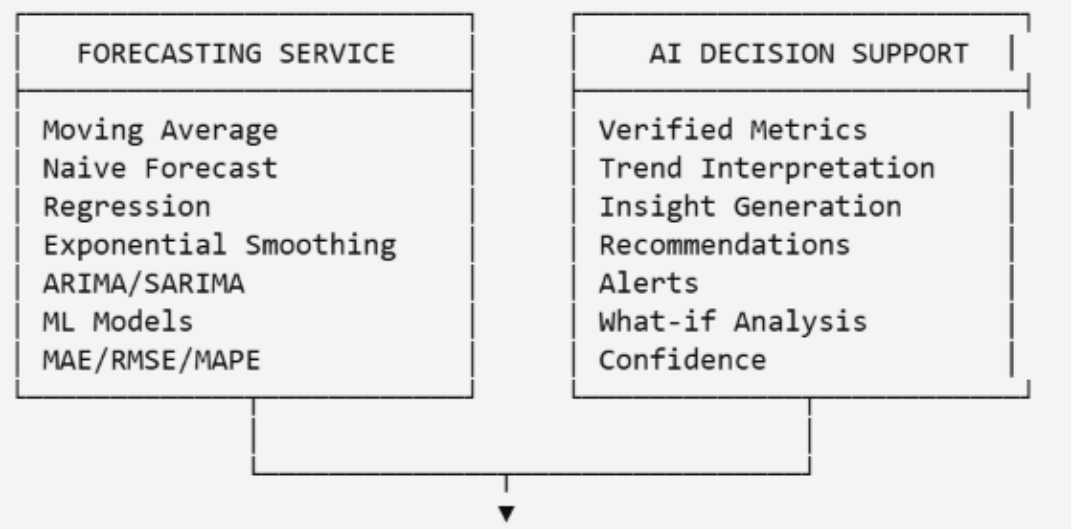
The AI layer must be based on validated analytical results rather than raw, unverified data.

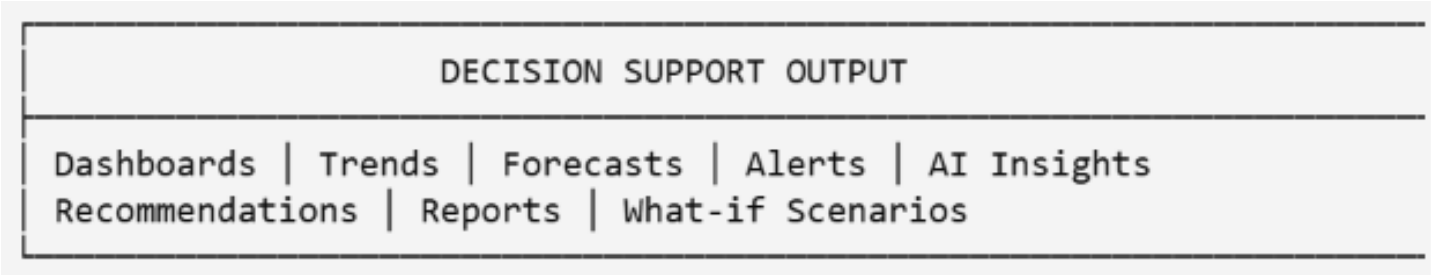
1. **Overall System Architecture**
2. **System Flow / Data Flow Diagram**
3. **Use Case Diagram**
4. **Entity Relationship Diagram (ERD)**
5. **AI-Driven Decision Support Workflow**

1. Overall System Architecture Diagram



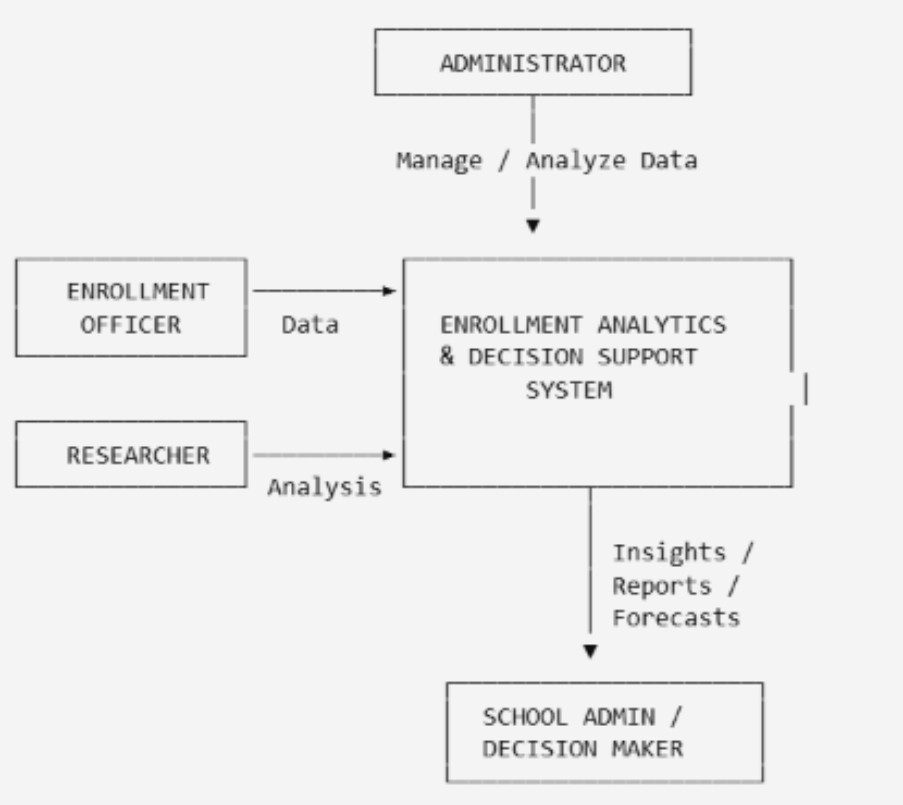

ENROLLMENT DB
Users
Academic Years
Semesters
Colleges
Departments
Programs
Enrollment Records
Forecasts
AI Insights
Audit Logs


FORECASTING SERVICE
Moving Average
Naive Forecast
Regression
Exponential Smoothing
ARIMA/SARIMA
ML Models
MAE/RMSE/MAPE
AI DECISION SUPPORT
Verified Metrics
Trend Interpretation
Insight Generation
Recommendations
Alerts
What-if Analysis
Confidence


DECISION SUPPORT OUTPUT
Dashboards | Trends | Forecasts | Alerts | AI Insights
Recommendations | Reports | What-if Scenarios

2. Data Flow Diagram — Level 0

This is suitable as your Context Diagram.



The external entities are:

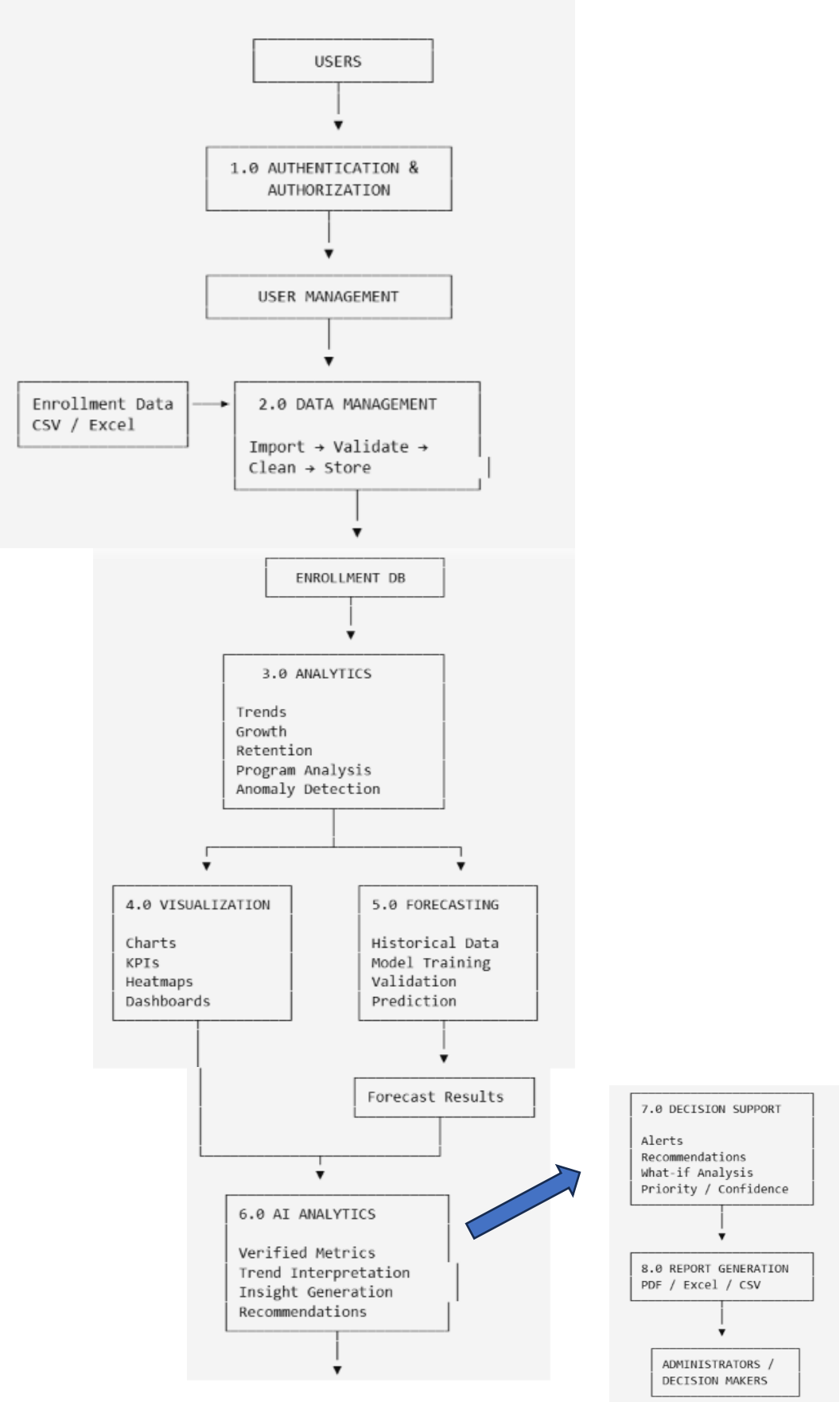
- **Administrator**
- **Enrollment Officer / Registrar**
- **School Administrator / Decision Maker**
- **Researcher / Analyst**

The central process is:

Visualization of Enrollment Dynamics Utilizing Data Analytics with AI-Driven Decision Support

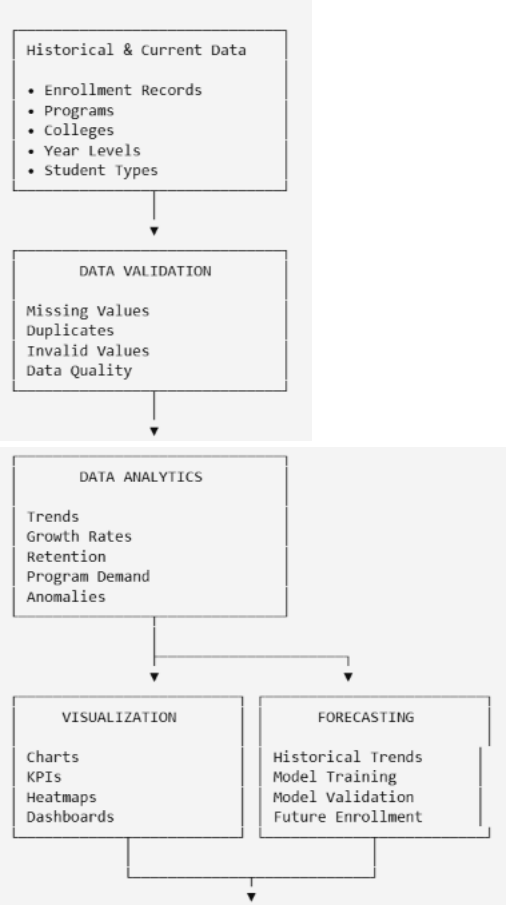
3. Data Flow Diagram — Level 1

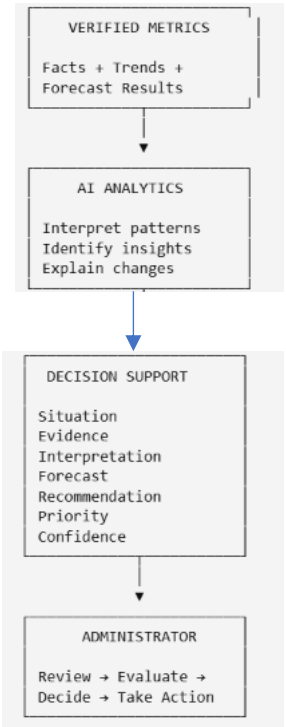
This gives a more detailed representation of how data moves through the system.



4. AI-Driven Decision Support Workflow

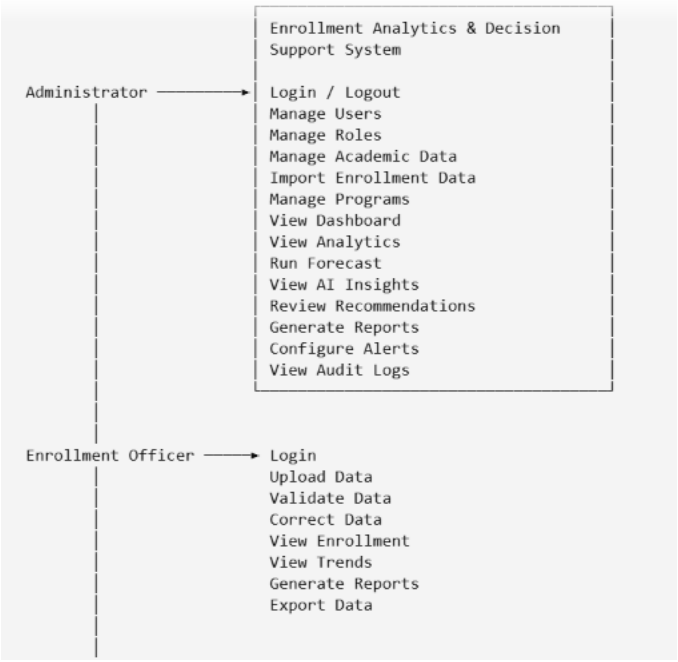
This is probably the **most important diagram** for your study because it shows how your system goes beyond ordinary visualization.





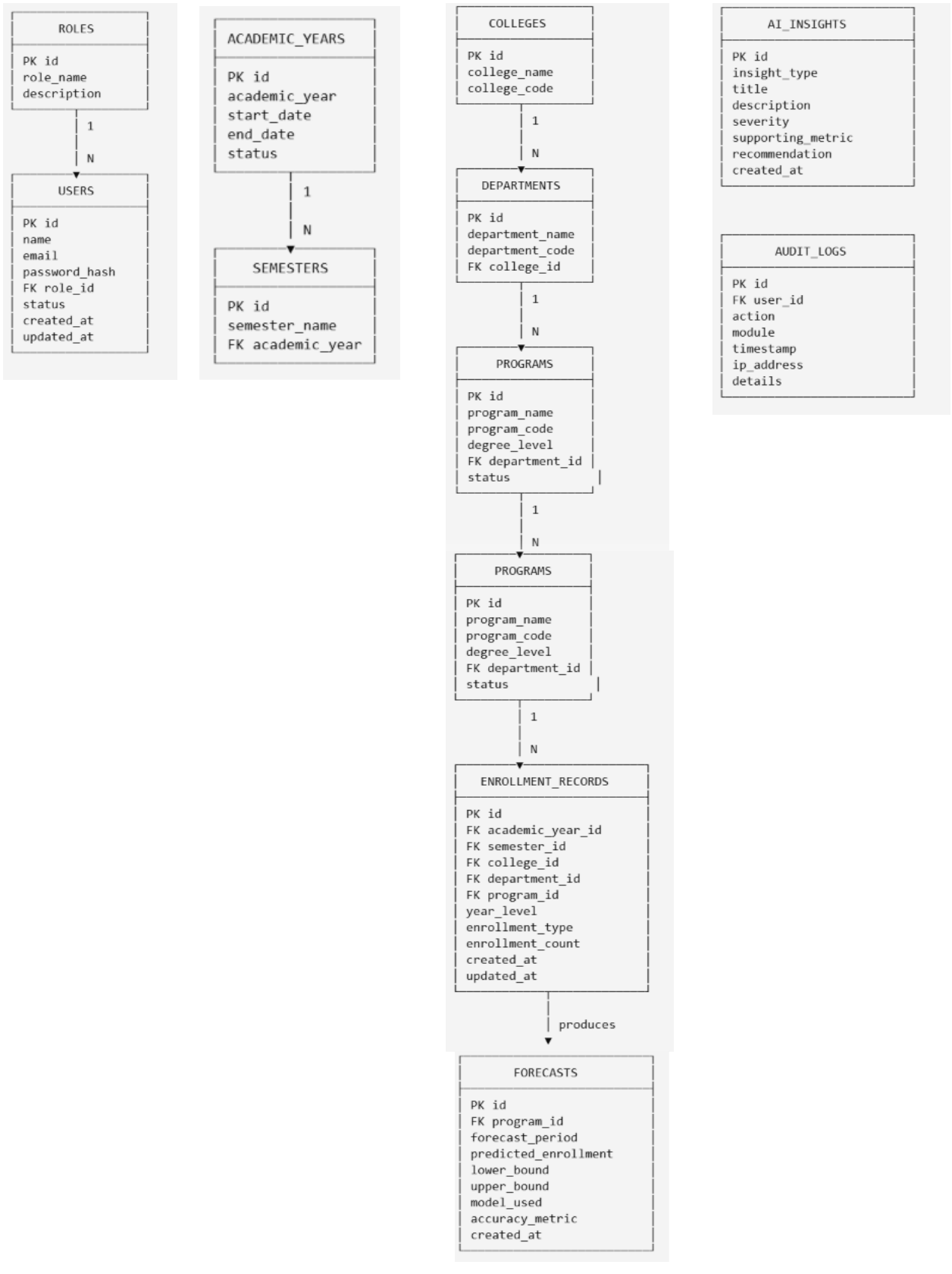
5. Use Case Diagram

You can structure your use-case diagram like this:



6. Entity Relationship Diagram (ERD)

For your database, I would improve the proposed structure slightly so that the relationships are clearer.



Recommended High-Level Thesis Figure

If you only need **one main diagram** for your proposal/Chapter 1, I recommend using this simplified conceptual architecture:

